

When Mass Security Screening Backfires:

A Signal Detection and Structural Causal Analysis

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Abstract: Mass screening programs for low-prevalence problems share a common signal detection structure that the National Academy of Sciences analyzed for polygraph screening through a single causal pathway: classification accuracy under Bayes' Rule. We extend this to a four-pathway structural causal framework incorporating classification, strategic adaptation, information, and resource reallocation effects, coupled to feedback dynamics that require σ -separation rather than d-separation and directed cyclic rather than acyclic graphs. The framework formalizes mass screening as performative prediction, addresses within-pathway aggregation through causal-consistency criteria, and demonstrates Bayesian uncertainty propagation. We analyze three security cases: the EU's proposed Chat Control 2.0 AI-scanning regulation; U.S. police polygraph programs, with a novel difference-in-differences analysis of nationally representative LEMAS data using negative binomial regression for the zero-inflated count outcome; and the iBorderCtrl facial micro-expression pilot. For Chat Control, all four pathways converge on net harm to child safety: classification backfires under rarity (99.4% false positives under generous assumptions); dedicated-offender evasion exceeds maximum casual-offender deterrence by a factor of ~ 46.6 ; the information pathway is architecturally absent; and false positive triage overwhelms investigative capacity. This convergence is robust across 27 normative-weighting cells. Police polygraph programs yield conflicting predictions — empirical evidence suggests modest harm while the simulation suggests possible modest benefit — and mechanism-level confession data are needed to sign the net effect. The recurring failure of such programs to demonstrate net benefit across domains suggests the need for a societal reckoning with this common, dangerous structure.

Keywords: mass screening; signal detection theory; structural causal models; false positives; screening policy

1. MOTIVATION

Mass screening of a whole population to catch a rare problem is among the most common policy instruments of the digital age, and it is proliferating. As communication, commerce, and administration move online, the same template recurs across otherwise unrelated domains: scanning all messages for child sexual abuse material (CSAM), all applicants for deception, all travelers for hostile intent, all patients for asymptomatic disease, all students for cheating, all transactions for fraud. Each is a signal detection problem of identical mathematical structure (Green & Swets, 1966), and each shares a failure mode that grows more consequential as the scanned population grows: under rarity, even an accurate test generates false positives that swamp the true ones, and where the screened condition cannot be independently verified and following up carries real costs, the net societal effect can be harm rather than benefit. Digitalization expands the reach of these programs and, with it, the stakes of getting their evaluation wrong. This paper supplies a framework for telling which such programs can plausibly succeed and which cannot.

The National Academy of Sciences' 2003 polygraph report, co-chaired by public statistician Stephen Fienberg, modeled estimated hypothetical outcomes in terms of true and false positives and negatives. Writing contemporaneously with the causal revolution, Fienberg implicitly focused on the proposed program's classification causal pathway. This single-pathway analysis applied Bayes' Rule to illustrate why polygraphing all National Lab scientists looking for spies would backfire whether the tests were set to minimize false positives or false negatives due to the accuracy-error trade-off, forcing "an unacceptable choice... between too many loyal employees falsely judged deceptive and too many major security threats left undetected" (NAS 2003, p. 6). This Bayesian backfire under the accuracy-error trade-off, and the overarching possibility of net harm from mass screenings of this structure, may be counter-intuitive due to the base rate bias (Piattelli-Palmarini, 1994), which often leads people to focus on high advertised *accuracy* numbers to the exclusion of

recognizing that the sheer number of false positives overwhelms the true positives when the problem of interest is rare.

Since Fienberg's analysis, causal modeling has revolutionized science (Pearl, 2009; Greenland, Pearl, & Robins, 1999; Glymour & Greenland, 2008). Such modeling reveals three additional pathways through which programs that share this common mathematical structure may affect outcomes:

1. Classification — test produces true and false positives and negatives (aka sensitivity and specificity)
2. Strategy — behavioral changes in response to the test (deterrence, evasion, gaming, and selection effects)
3. Information — knowledge generated or spread through the screening process itself (elicitation, bogus pipeline, suggestion, and some forms of iatrogenesis)
4. Resource reallocation — both quantitative (zero-sum finite capacity) and qualitative (framing effects)

Thus, we cannot estimate such programs' net effects from classification pathway effect estimation alone. Even if the test backfires via this pathway due to the accuracy-error tradeoff under common conditions of (1) rarity of the problem or condition of interest, (2) persistent inferential uncertainty about classification, and (3) secondary screening harms, the strategy and information pathways may still dominate with benefits. Or, along with resource reallocation effects, the false positive backfire along the classification and other pathways may amplify net harm.

Three case studies demonstrate the utility of this new framework for estimating the causal effects of mass screenings for low-prevalence problems, ordered by depth of treatment. Chat Control, the

common colloquial name for the proposed EU Child Sexual Abuse Regulation (CSAR) on preventing and combating child sexual abuse, receives the most detailed analysis here. The Commission, under Home Affairs Commissioner Ylva Johansson, proposed mandatory AI scanning of digital communications in May 2022 (COM(2022) 209 final), aiming to catch the transmission of CSAM and grooming. After three years of contested negotiations, the Council reached a position in November 2025 (Council of the European Union 2025) that retreated from mandatory scanning of end-to-end encrypted communications (e.g., WhatsApp and Signal). Breyer (2025) documents the legislative timeline.

Until 3 April 2026, voluntary scanning operated on services encrypted in transit but unencrypted to the service provider (e.g., Gmail), using hash-matching of known CSAM. That regime ended when the temporary ePrivacy derogation authorizing voluntary detection expired on 3 April 2026 without renewal (Breyer 2025); on 9 July 2026, the European Parliament fell short of the absolute majority required to reject the Council's fast-tracked reinstatement of the derogation, while adopting amendments explicitly excluding end-to-end encrypted services; at the time of writing, the amended file is back before the Council and the reinstated derogation is not yet in force.

Hash-matching of known CSAM operates under fundamentally different logic: it matches against confirmed material rather than classifying unknown content, so a flag can be verified rather than merely inferred. This dissolves the persistent inferential uncertainty that deepens the mandatory proposal's false positive burden.

Critics have raised two strategic adaptive concerns about perceptual hashing (edge-detection-based content fingerprinting; Marr & Hildreth, 1979): evasion (small image modifications producing large hash value changes, increasing false negatives) and sabotage (adversarial flooding to increase false positives) (Troncoso et al., 2024). These are possibilities, but they do not resurrect the base rate problem. The “security by obscurity” objection — that any system depending on hash secrecy is

cryptographically unsound — imports a principle from classical cryptography that pertains to algorithm secrecy, where it is rightly criticized, and misapplies it to hash list secrecy, which serves a different function: protecting against evasion and sabotage rather than providing cryptographic security. Finally, hash matches require physical corroboration — when investigators collect devices, the evidence either holds or it does not — limiting the scope for authoritarian abuse through hash manipulation compared to purely algorithmic classification.

This framework applies specifically to programs that must infer unknown status — e.g., novel abusive content, deceptive intent, or threat classification — from visual, physiological, or behavioral input with probabilistic rather than one-to-one signal versus noise correspondences. Detection of known, pre-specified signals operates under fundamentally different statistical logic, as the hash-matching case illustrates. That distinction defines the scope of what follows: all three case studies involve the former.

The analysis here concerns the previously proposed mandatory AI screening of all digital telecommunications. This analysis matters because member states retain discretion over implementation. Trilogue negotiations over the permanent framework (CC 2.0) remain unresolved: the fifth scheduled round, on 29 June 2026, ended without agreement over suspicionless detection, and a sixth round is expected in September 2026 under the Irish Presidency. The Council’s negotiated text dropped explicit mandatory-detection orders in favor of “risk-mitigation” obligations, but critics argue those obligations would compel scanning in practice, and earlier mandatory-scanning language can be reintroduced in future revisions. The scenario analyzed here is therefore the live policy question, not a hypothetical one. The current sociopolitical stakes of estimating the net effects of this program are thus the highest of the considered cases. This analysis suggests harm may dominate along every causal pathway, an important finding for child safety.

Police polygraph programs embody a different causal structure: strategy and information effects may dominate due to deterrence and bogus pipeline phenomena, producing conflicting predictions. This analysis may generalize to mass security screenings with strategic selection effects and adaptive human interviewing components.

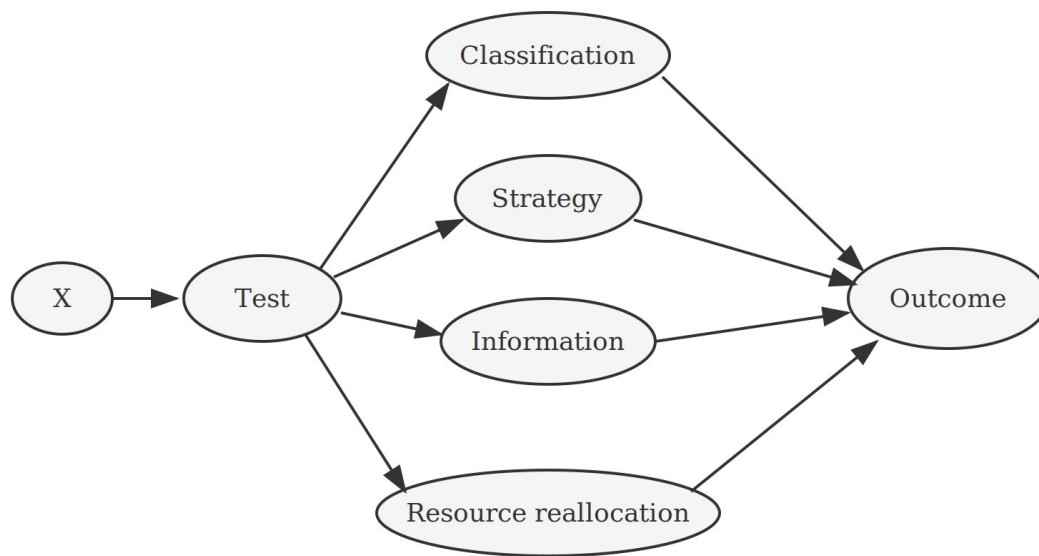
iBorderCtrl receives the most abbreviated treatment; its automated kiosk format has the form of interrogation-based coupling without the substance — lacking adaptive human elicitation — and sustained scientific and civil-society critique prevented deployment beyond the pilot phase — though the program's own performance evidence was never made public (see §7). This gradient — from strong possible strategic and information effects in polygraph programs, to attenuated coupling in iBorderCtrl, to none in Chat Control — logically predicts which programs may compensate for classification pathway backfire through other pathways, and which cannot. Yet, the empirical results are sobering: police polygraph programs, like Chat Control and iBorderCtrl, may backfire.

Importantly, the framework applies specifically to *mass* screening regimes, where tests are applied population-wide to detect rare conditions. Targeted *investigative* use of similar tools operates under an entirely different statistical structure: base rates are substantially higher and screening occurs after suspicion has already narrowed the population. The causal effects of mass versus targeted screenings may be expected to differ substantially. This application considers only causality in mass screening programs.

Security programs often present special validation challenges that make inferential uncertainty particularly persistent and secondary screening harms particularly grave. The condition being detected — deceptive intent, future criminal behavior, threat status — seldom admits determinative forensic, witness, or confessional evidence that could conclusively confirm a given classification;

and because behavior is endogenous to the screening itself, ground truth is unobservable not merely in practice but in principle: the act of screening alters the very outcome any validation would measure. Errors therefore compound silently, and harms to falsely flagged individuals accumulate without corrective feedback. Future work will extend this framework to other domains including medicine, where ground truth is often eventually observable and validation is more tractable.

2. MULTIPLE CAUSAL PATHWAYS



DAG: Multiple Causal Pathways in Mass Screening

Figure 1. The DAG shows multiple causal pathways beyond the typical classification focus are relevant to net program effects.

The four pathways from Test to Outcome:

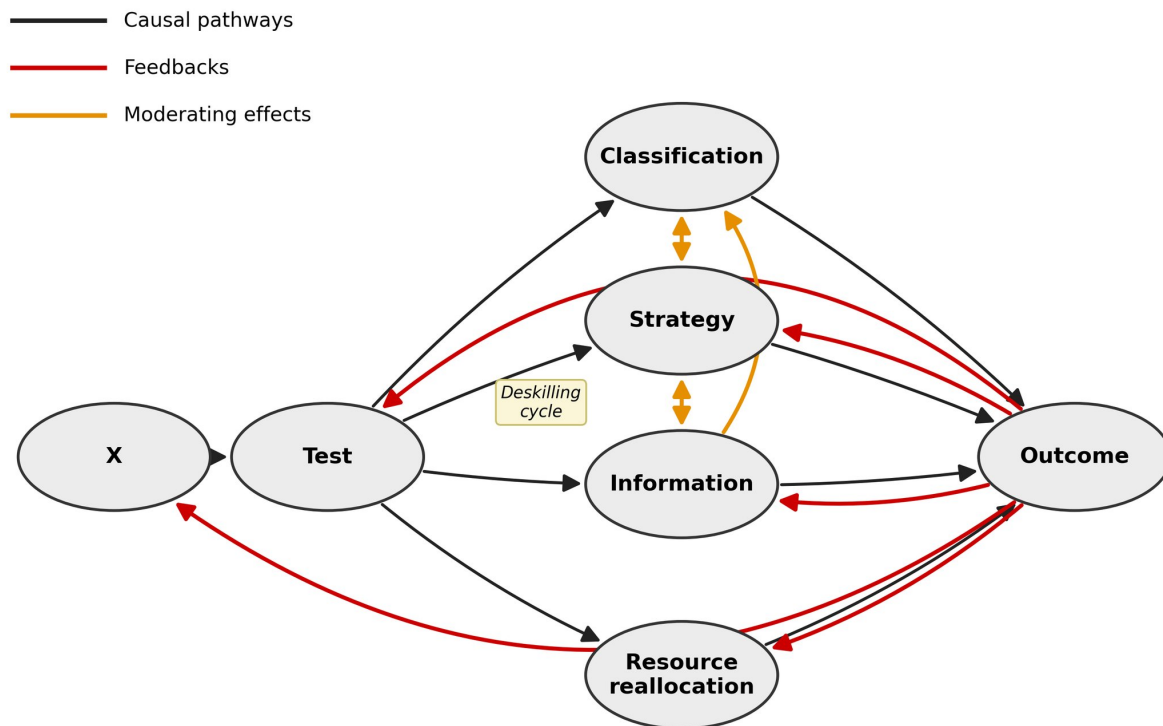
CLASSIFICATION: True Positives, False Positives, True Negatives, False Negatives

STRATEGIC BEHAVIOR: Deterrence, Evasion/Sabotage, Perverse Incentives, Selection

INFORMATION EFFECTS: Elicitation, Suggestion, Iatrogenesis

RESOURCE ALLOCATION: Quantitative (zero-sum), Qualitative (framing)

3. FEEDBACK LOOPS IN EQUILIBRIUM



DCG: Multiple Causal Pathways with Feedbacks in Mass Screening

Figure 2. The DCG extends the acyclic model to a directed cyclic graph, incorporating the feedback dynamics (red arrows) — including the deskilling cycle between strategy and information pathways

— that motivate the equilibrium modeling approach. It also incorporates moderating effects (orange arrows). For representation and reasoning in cyclic causal models see Ahsan, Arbour & Zheleva (2022). Because the DCG contains feedback loops, conditional independence reasoning requires σ -separation rather than d-separation (Forré & Mooij, 2018) — the cyclic analogue that accounts for equilibrium dependencies.

3.1 Feedback loops

Feedback loops (Outcome $\rightarrow$ earlier nodes):

- Outcome $\rightarrow$ Test: Programs that appear to “work” get expanded; programs that appear to fail get cut — or paradoxically expanded
- Outcome $\rightarrow$ Strategy: Observed outcomes change behavioral responses
- Outcome $\rightarrow$ Information: What’s learned affects future elicitation
- Outcome $\rightarrow$ Resource reallocation: Results drive resource decisions
- Outcome $\rightarrow$ X (Selection): Program outcomes affect who enters the screening pool

3.2 Moderating effects

Moderating effects (pathway interactions):

- Strategy $\leftrightarrow$ Classification: Gaming/evasion degrades effective test accuracy.
- Information $\rightarrow$ Classification: Confessions affect apparent test performance metrics
- Strategy $\leftrightarrow$ Information: How targets behave strategically shapes what information the screening process generates, and vice-versa. In polygraph screening, the interrogation process (information pathway) elicits confessions only from applicants who believe the test

works and choose to address perceived detection — a strategic calculation, cognitive error, or emotional response. What interrogators then learn from confession patterns feeds back to change interrogation strategy, while applicants who learn what the polygraph interview focuses on or is like can prepare accordingly. In Chat Control, this coupling is largely absent — automated scanning generates no interactive information, and targets' strategic evasion may produce no usable intelligence feedback. In other contexts, this coupling may manifest as a deskilling cycle: strategic behavioral adaptation (reduced vigilance when AI is present) produces cumulative skill degradation (an information effect), which reinforces the strategic adaptation (increased AI reliance). This cycle is a cross-domain concern. In security screening, it may manifest as analyst over-reliance on algorithmic flags (confirmation bias); in medical screening, as reduced clinical perspicacity when AI is present. The strength of Strategy ↔ Information coupling may be a key structural difference between screening programs that benefit from non-classification pathways and those that do not.

3.3 Erroneous beliefs and ineffective programs

People can erroneously believe programs work and advocate for their expansion:

- As in medical screening more broadly (see §8), cancer screening survivors who were overdiagnosed with non-progressive cancers may wrongly believe their lives were saved; indeed, there are tremendous psychological incentives to believe so. As a consequence, cancer screening programs are hugely popular despite decades of critical advocacy for informed consent about their possible harms.

- Organized advocacy for Chat Control has included funding structures and litigation strategies suggesting organizational rather than grassroots origin (Heat Initiative/Hopewell Fund; Biddle, 2023), complicating interpretation of apparent public support.
- Polygraph and other mass screening programs, once entrenched, are notoriously difficult to gut regardless of their scientific evidentiary basis: the “lie detector” mystique persists despite decades of devastating scientific critique.

This creates positive feedback loops on perceived rather than actual effectiveness—programs expand based on testimonials and political pressure (“but we have to do *something*”), rather than on scientific evidentiary evaluation.

Surveillance-induced guilt without confession

The Information pathway is more complex than simple confession elicitation. Flagged individuals who believe they were caught may act guilty without confessing to authorities, destroying evidence and potentially even themselves. Such guilty behavior may carry some social benefits outside formal criminal justice channels. For innocent flagged individuals, however, such surveillance-induced psychological repercussions represent pure harm with no offsetting benefit of any kind.

3.4 Cross-pathway dynamics

The Classification↔Strategy and Classification↔Information moderating effects are addressed in §3.2; the focus here is the Strategy↔Information coupling, which is the most consequential cross-pathway dynamic for comparative security screening program analysis.

The Strategy and Information pathways are not independent channels from Test to Outcome.

Rather, they are coupled: strategic behavior moderates what information the screening process can generate, and information generated through screening feeds back to reshape strategic behavior.

This coupling is represented by the Strategy ↔ Information arrows.

Note that selection effects (e.g., applicants refraining or withdrawing before reaching the interrogation stage of police polygraphs) represent strategy pathway contributions that bypass the information pathway entirely, further complicating decomposition from outcome data alone.

Again, iBorderCtrl occupies an intermediate position: its kiosk interview format has the appearance of an interrogation — potentially activating the bogus pipeline — but the automation strips out the adaptive human elicitation that may make polygraph interrogation effective along the information pathway. It has the form of Strategy ↔ Information coupling without the substance.

Overall, the strength of this coupling explains why polygraph programs may net benefit through non-classification pathways while Chat Control cannot: the Strategy ↔ Information coupling that could compensate for classification pathway failure is architecturally absent.

3.5 Limitations

The parameter estimates are transported across contexts (lab-based bogus pipeline effects applied to field screening; street-crime deterrence estimates applied to applicant screening). This transportability problem is a specific instance of the validation problem that structures the proposed cross-domain framework: the bogus pipeline effect was measurable in lab studies because ignorant subjects confessed. But in security screening contexts, such confessions may be less likely due to different conditions including higher stakes — meaning neither the parameter estimate nor the

screening program's classifications can be validated against an observable answer sheet. The transportability uncertainty is unquantified and probably much larger than the within-study sampling uncertainty reported here.

The closest formal tool for cross-context inference is transportability: Hünemann and Bareinboim (2025) encode source–target differences as selection nodes and use do-calculus to decide whether an effect identified in one population is identifiable in another. But transportability presupposes that the source effect is itself identifiable — that an answer sheet exists to port. The security validation problem is prior to that question: where the construct is constitutively unobservable, there is no source-population answer sheet for any amount of do-calculus to transport, and a finding that the effect ports would signal the model is misspecified.

Gelman and Carpenter (2020) demonstrated that even within a single study and population, treating classification parameters as known rather than uncertain can substantially alter substantive conclusions when conditions are rare. Security applications compound this problem by additionally transporting across contexts which are likely to be more heterogeneous in this domain than in others (lab and field). Functional forms are assumed (largely linear) without adequate theoretical justification. Future work should develop multi-level versions of this paper's structural equations with within-pathway heterogeneity, apply Bayesian uncertainty propagation (Gelman and Carpenter 2020) to all parameters, conduct formal sensitivity analysis across functional forms, and — where validation data exists (medical applications with observable ground truth) — test whether the single-level model produces valid causal predictions. The current framework is best understood as a qualitative tool for identifying which pathways matter, how they interact, and what the likely signs are along each pathway and in net, rather than a quantitative tool for estimating effect sizes.

3.6 The Aggregation Problem Revisited

The four-pathway framework was motivated by a critique of Fienberg et al.'s single-pathway analysis in the National Academy of Sciences polygraph report: collapsing all possible causal mechanisms into a single classification accuracy question neglects causal structure and produces misleading policy conclusions. At best, it produces only misleadingly overconfident conclusions without causal inferential basis. At worst, it produces completely wrong conclusions.

At the level of paper organization, the strategy pathway is examined before the Bayesian backfire analysis (§5.1) because, for Chat Control, it carries the strongest compensating argument proponents can make: with the information pathway architecturally absent, deterrence is the only non-classification channel that could offset classification backfire. If it did, the net analysis could still support implementation. Establishing across the full parameter space that it cannot lets the Bayesian results that follow be read with that compensating argument already foreclosed, so their practical weight is clearer.

At the level of scientific inference, Fienberg's analysis illustrates what Cartwright (1989) calls a fundamental constraint: no causes in, no causes out. An analysis that encodes only the classification pathway cannot produce conclusions about strategy, information, or resource effects — not because those pathways are absent, but because they were never specified. The four-pathway causal framework is a direct response to this constraint.

While this critique is valid, the proposed causal framework risks replicating the error at a different level of aggregation. Fienberg's aggregation error was a vertical one of misspecification: ignoring three of four causal pathways, implicitly assuming a *homo economicus* world in which a clean diagnostic test meets rational subjects and neutral institutions — a construction that neglects

complex system dynamics including strategic adaptation, belief effects, and resource implications (Simon, 1955; Tversky & Kahneman, 1974; Herley, 2014; see also §3.2–3.5).

The risk here is horizontal: collapsing within-pathway heterogeneity into single aggregate parameters. The strategy pathway’s “net deterrence effect” combines casual offenders who are deterred with sophisticated offenders who evade — two subpopulations with potentially opposite causal contributions. Aggregating across these subpopulations may produce causal parameters that do not correspond to any interventionally manipulable quantity. Rubenstein et al. (2017) formalize this as causal consistency: an aggregate causal model is consistent with a fine-grained model only under specific conditions the analyst must establish rather than assume.

As Rubenstein et al. and others explain, this is the cholesterol problem: LDL and HDL have opposite effects on cardiovascular risk, so aggregating them into “total cholesterol” does not just lose information — it creates a variable that is causally incoherent as a target of intervention. “Net deterrence” in the strategy pathway may be similarly incoherent.

The same problem also applies to the resource reallocation pathway: aggregating national police forensic investigators and specialist prosecutors into a single ‘capacity’ figure obscures that these two groups likely have divergent elasticities. The first, forensic investigative capacity, can theoretically be expanded with sufficient resources by expanding funding to hire more police investigators. The second, specialist prosecutor pipelines, may be more supply-constrained by market and professional factors regardless of available funding (Hartmann, expert interview, May 2025). Figures attributed to Hartmann derive from a documented expert consultation with Markus Hartmann (Head of ZAC NRW, Generalstaatsanwaltschaft Köln; structured interview 21 May 2025, follow-up correspondence March 2026), who endorsed the project in a letter of support (ZAC NRW, 8 August 2025).

In medical screening — where we can often eventually observe ground truth (e.g., in cancer-specific mortality in aggregate terms, or histopathology/lab reports in cases on the poles of the pathology spectrum) — we can assess whether aggregate parameters produce valid causal predictions. In security screening, such measures are often far harder or impossible to obtain, and the validation problem is not merely practical but structural. For present latent states like deceptive intent, determinative corroborating evidence is seldom available; for future outcomes like misconduct or criminal behavior, the target is unobservable not merely in practice but in principle, because behavior is endogenous to the screening itself — the act of screening alters the very outcome any validation would measure.

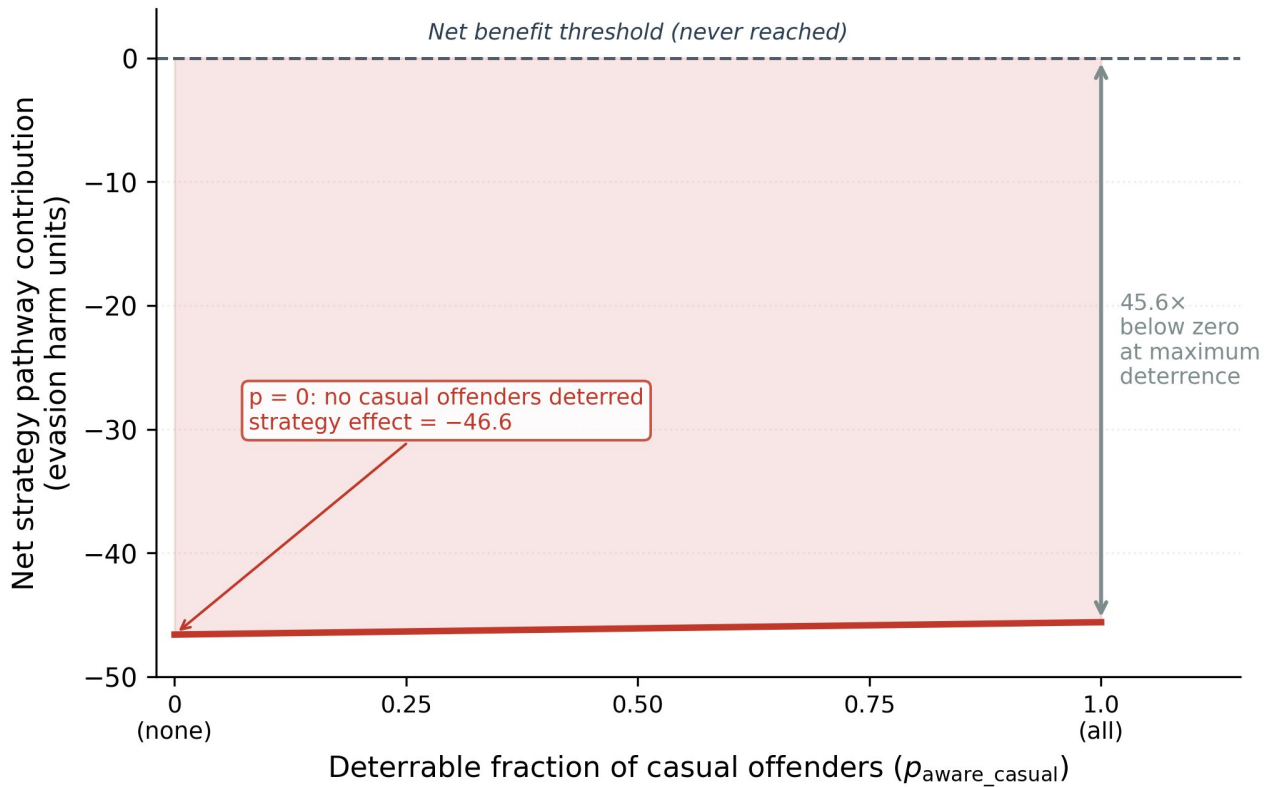


Figure 3. *Chat Control Strategy Pathway: Sensitivity to Casual-Offender Deterrability.* Net strategy pathway contribution as a function of the deterrable fraction of casual offenders ($p_{\text{aware_casual}} \in [0, 1]$). Even assuming all casual offenders are deterrable (right edge), dedicated-offender evasion exceeds casual-offender deterrence by a factor of 46.6. The line never approaches zero. The shallow slope itself is substantively meaningful: the entire range of uncertainty about casual-offender deterrability moves the outcome by less than 2% of the evasion term. The compensating subpopulation argument is foreclosed across the full parameter space.

Perdomo et al.’s foundational work on performative prediction (2020) formalizes this: once a predictive model is deployed, the population it describes begins adapting to it, shifting the very distribution the model was trained on. Hardt et al. (2016) show this is especially acute under strategic classification — when subjects actively game the system. In security screening, this feedback is not a nuisance — it is constitutive of the system’s causal structure. In medical diagnosis, the prediction “this patient has cancer” does not change whether the patient has cancer. In security

screening, the prediction “this person intends to deceive” changes the conditions under which deception will or will not occur — through deterrence, evasion, strategic adaptation, and the psychological effects of being screened. Security screening is not applied statistics on a fixed population; it is an intervention that reshapes the population it claims to characterize. The minority report problem is not solvable by improving classifier accuracy, because the act of prediction changes the future being predicted.

Any framework that requires knowing a priori which subgroups matter and how their causal contributions aggregate faces an epistemological ceiling in security applications that does not apply to medical counterparts. Cross-domain validation will need to proceed from medical or other applications with better-solved validation problems, back toward security applications, rather than the reverse. The current paper is a step toward that work.

The methodological response to the horizontal aggregation problem is a decision rule: *disaggregate a pathway when and only when proponents of a program can invoke a compensating subpopulation argument* — i.e., when one subpopulation’s positive pathway contribution might plausibly offset another’s negative contribution in the aggregate. For Chat Control, this argument is live: proponents could claim that casual-offender deterrence compensates for dedicated-offender evasion in the strategy pathway. Disaggregation directly answers that claim.

Both pathways have been disaggregated and implemented: the strategy pathway separates casual offenders into truly unaware and vaguely aware subpopulations alongside dedicated offenders, with sensitivity analysis over the deterrable fraction (`disaggregated_strategy_v2.py`); the resource pathway separates triage-layer and specialist-layer capacity, showing the specialist bottleneck is a binding constraint regardless of triage scaling (`disaggregated_resource_sim.py`). That is: the bottleneck presents at the first level in both current practice and proposed possible infrastructure;

but even if that resource problem were overcome, another, more intractable one would then present. The overload result appears robust to possible structural changes.

Both files are at <https://github.com/verawilde/equilibrium-screening/tree/main/security>; results are in §5.3–5.4 and Supporting Information S1. Under maximally generous assumptions, dedicated-offender evasion still dominates casual-offender deterrence by a factor of 46.6. The net harm result is robust to assumptions about deterrability.

For polygraph programs and iBorderCtrl, no analogous compensating subpopulation argument is central to the proponent case, so disaggregation is not required there.

4. STRUCTURAL EQUATIONS

Following Besserve and Schölkopf 2022's notation, let the system reach equilibrium x^* satisfying self-consistent equations.

4.1 Variables

- Population (X): Number of individuals screened
- Test intensity (T): Screening coverage/thoroughness, range [0,1]
- Classification output (C):

$$TP = T \cdot X \cdot \pi \cdot \text{sensitivity (true cases caught)}$$

$$FP = T \cdot X \cdot (1 - \pi) \cdot (1 - \text{specificity}) \text{ (innocents flagged)}$$

$$FN = T \cdot X \cdot \pi \cdot (1 - \text{sensitivity}) \text{ (true cases missed)}$$

$$TN = T \cdot X \cdot (1 - \pi) \cdot \text{specificity (innocents correctly cleared)}$$

- Strategic behavior (S): Net deterrence effect (positive = deterred, negative = evasion)

- Information yield (I): net information effect of the screening process (positive = actionable intelligence and valid elicitation; negative = iatrogenic harm — false confessions, suggestion, surveillance-induced guilt in the innocent), range $[-X, X]$.
- Resource allocation (R): Proportion of investigative resources on mass vs targeted, range $[0,1]$
- Outcome (Y): Security/integrity measure (higher = better)
- Base rate (π): True prevalence of target condition, range $[0,1]$

4.2 Structural Assignments

Classification pathway: $C := f_C(T, X, \pi, \epsilon_C) = T \cdot X \cdot [\pi \cdot \text{sensitivity} + (1-\pi) \cdot (1-\text{specificity})]$

Where sensitivity and specificity are test operating characteristics. Under rarity ($\pi \rightarrow 0$), C is dominated by false positives.

Strategy pathway: $S := f_S(T, C, \theta_S) = \theta_{\text{deterrence}} \cdot T \cdot g_{\text{awareness}}(T) - \theta_{\text{evasion}} \cdot h_{\text{sophistication}}(C)$

Where $g_{\text{awareness}}(T)$ captures how visible/credible the screening program is, $h_{\text{sophistication}}(C)$ captures learning to evade based on classification feedback, and θ parameters capture population characteristics.

Literature on deterrence effects:

- Pratt et al. (2006) meta-analysis: Certainty of apprehension matters more than severity of punishment

- Nagin (2013): “It is the certainty of apprehension, not the severity of the ensuing consequences, that is the more effective deterrent”
- Effect sizes for deterrence are generally modest and uncertain ($r \sim 0.1-0.2$, with wide intervals), and depend heavily on perceived certainty
- Severity of punishment shows effects close to zero with intervals spanning both directions (Doob and Webster 2003)

Information pathway: $I := f_I(T, S, \theta_I, \lambda) = \theta_{\text{confession}} \cdot T \cdot (1 - S/X) \cdot \text{bogus_pipeline_effect} \cdot \lambda$, where $\lambda \in [-1, 1]$ is a transport/sign coefficient: $\lambda \rightarrow 1$ if the laboratory bogus-pipeline effect transports to field screening as a benefit, $\lambda \rightarrow 0$ if it fails to transport, and $\lambda < 0$ if it inverts in the field (adverse selection — conscientious over-disclosure screened out, or low-reactivity deceivers selected in).

The bogus pipeline effect captures that people confess more when they believe a test works, regardless of whether it actually does.

Literature on bogus pipeline:

- Roese and Jamieson (1993) meta-analysis of 31 studies: $d = .41$, 95% compatibility interval (CI) approximately .25-.57 (positive sign, moderate effect magnitude, meaningful uncertainty)
- Effect is larger when participants must “guess” what the machine detected
- Works by shifting motivation from self-enhancement to self-protection (fear of being caught lying)
- Note: Effect sizes from this literature should be interpreted cautiously given replication concerns in psychology generally

Resource reallocation pathway: $R := f_R(C, I, \theta_R) = \text{base_allocation} + \theta_{\text{false_positive_burden}} \cdot C/\text{capacity} - \theta_{\text{tip_efficiency}} \cdot I$

More false positives drain resources; good tips improve allocation efficiency.

Outcome (equilibrium): $Y^* := f_Y(TP, FP, FN, S, I, R, \pi, X) = \theta_{TP} \cdot TP - \theta_{FP} \cdot FP - \theta_{FN} \cdot FN$

[classification contribution: benefit of cases caught, harm of innocents flagged, harm of cases missed] + $\theta_S \cdot S$ [strategy contribution] + $\theta_I \cdot I$ [information contribution] –

$\theta_R \cdot R \cdot \text{opportunity_cost}$ [resource reallocation cost]

4.3 Feedback Loops

Outcome → Test feedback: $T_{\{t+1\}} := f_{\text{feedback}}(T_t, Y_t, \text{perceived_}Y_t, \text{political_pressure})$

Note: perceived_Y may diverge from actual Y due to false positive testimonials, organized advocacy, institutional inertia and sunk costs, and difficulty measuring counterfactual outcomes.

Outcome → Selection feedback: $X_{\{t+1\}} := f_{\text{selection}}(X_t, Y_t, \text{program_expansion})$

Strategy ↔ Classification moderation: $\text{effective_sensitivity} := \text{base_sensitivity} \cdot (1 - \text{evasion_factor}(S))$
 $\text{effective_specificity} := \text{base_specificity} \cdot (1 - \text{gaming_factor}(S))$

Strategic evasion and gaming degrade test performance.

5. APPLICATION 1: CHAT CONTROL 2.0

EU Chat Control 2.0 — the Commission’s May 2022 proposal (COM(2022) 209 final) for mandatory AI scanning of all digital communications — is the subject of this analysis. As noted in

§1, the voluntary (CC 1.0) regime used hash-matching of known CSAM and operated under different statistical logic; the concerns analyzed here apply specifically to the mandatory AI classification proposal.

5.1 Bayesian baseline: frequency-format estimates under favorable assumptions

Following Fienberg et al.'s approach of applying generous assumptions to illustrate a proposed mass screening for a low-prevalence problem's beyond-best-case scenario, we apply Bayes' Rule to Chat Control using parameters intentionally favorable to proponents: a true positive rate of 90%, a false positive rate of 16%, and a base rate of 1 CSAM message per 1,000 messages scanned. Scaling to 10 billion messages — a fraction of daily European digital communications — Tables 1a and 1b show results under two operating thresholds.

The 90% true-positive rate is generous even relative to optimized tools operating on pre-filtered evidence: Hartmann (expert interview, 21 May 2025) reports 5–7% error rates for AI tools on pre-filtered seized evidence, reflecting in-domain performance on a targeted, high-prevalence subpopulation. The content-classification literature consistently finds detectors shedding roughly 30 points of macro-F1 under distribution shift (Yin & Zubiaga 2021), with validated models collapsing from above 90% to below 50% F1 out-of-domain (Arango et al. 2020); on a representative random sample of real traffic — the closest analog to unfiltered mass screening — the best detectors reach only ~47% F1, which the authors judge unfeasible for deployment (Tonneau et al. 2025). The assumed 90% sensitivity is therefore generous on two counts, distributional and base-rate, so the central result is best read as a lower bound on the false positive burden.

Table 1a. *Hypothetical Estimated Outcomes of Chat Control in High-Sensitivity Mode.* The first of two scenarios with different operating thresholds showing the same structural signal detection

Chat Control result	Non-abusive (innocent)	Abusive (CSAM)	Total
Flagged content	1,598,000,000	9,000,000	1,607,000,000
Not flagged content	8,392,000,000	1,000,000	8,393,000,000
Total	9,990,000,000	10,000,000	10,000,000,000

problem, illustrating the accuracy-error trade-off when scanning 10 billion messages with a 90% accurate test. This scenario maximizes detection but generates 1.6 billion false positives (99.4% of flags innocent).

Under these unrealistically favorable assumptions, both scenarios reveal the same structural problem from opposite directions: Table 1a maximizes detection but generates 1.6 billion false positives (99.4% of flags are innocent); Table 1b minimizes false positives but leaves 80% of real CSAM undetected. There is no threshold that avoids both failures simultaneously. This is not a calibration problem solvable by adjusting the threshold; it is a mathematical consequence of screening for a rare condition with an imperfect test.

Chat Control result	Non-abusive (innocent)	Abusive (CSAM)	Total
Flagged content	39,000,000	2,000,000	41,000,000
Not flagged content	9,951,000,000	8,000,000	9,959,000,000
Total	9,990,000,000	10,000,000	10,000,000,000

Table 1b. *Hypothetical Estimated Outcomes of Chat Control in High-Specificity Mode.* The second scenario with the test set to minimize false positives leaves 80% of abuse undetected.

Under the high-sensitivity scenario (Table 1a), a flagged message has a 99.4% probability of being innocent — approximately 178 innocent people are flagged for every genuine CSAM message

detected. Under the high-specificity scenario (Table 1b), the false positive burden falls, but 80% of CSAM evades detection entirely, meaning 8 million abusive messages per 10 billion reach their recipients undetected. There is no threshold at which both failures are avoided simultaneously. This is the accuracy-error trade-off under rarity: not a calibration problem, but a mathematical consequence of the structure of the signal detection problem itself — which is a consequence of the structure of reality and not something better technology can fix.

5.2 Relaxing assumptions: Bayesian uncertainty propagation

The baseline above treats accuracy (90% true positive rate, 84% true negative rate) and base rate (1/1,000) as known. However, they are substantially uncertain. Gelman and Carpenter (2020) demonstrate that treating classification parameters as fixed rather than uncertain can substantially alter substantive conclusions when the target condition is rare. These are the conditions under which Chat Control would operate.

The base rate of CSAM in digital communications is unverified and contested. Mass scanning proposal proponents have strong incentives to inflate it (making the program appear more beneficial) and to inflate novel scanning methods' accuracy (making detection appear more reliable). Independent verification of either parameter is unavailable; the European Commission's impact assessment (European Commission 2023) relied on industry-provided figures from Microsoft and Thorn, with no independent verification. In this as in other realms, non-transparency blocks neutral evidentiary evaluation.

Propagating this uncertainty: if the true base rate is 1/10,000 rather than 1/1,000 — plausible given that most CSAM circulates in dedicated encrypted channels rather than general communications —

the false positive-to-true-positive ratio worsens tenfold (from $\sim 178:1$ to $\sim 1,778:1$), and the true positive rate of the system falls below 0.05%. If accuracy is 80% rather than 90% — still generous — the ratio of false to true positives worsens further. Subsequent operational reporting is consistent with these directions: the Commission's implementation report on the voluntary regime (European Commission 2025) records error rates of 13–20% for classifiers targeting unknown material even on already-filtered review pools — figures that are false discovery rates on enriched samples rather than false positive rates in the specificity sense, and therefore cannot be extrapolated to population-scale scanning — and suggests the true prevalence of illegal content in scanned communications is orders of magnitude below the $1/1,000$ assumed here. The combination of lower base rate and lower accuracy produces a system that functions almost entirely as a mass surveillance apparatus with negligible child protection benefit.

This sensitivity to parameter uncertainty is not incidental; it is structural. Under conditions of rarity, small changes in base rate assumptions produce large changes in predicted outcomes. Any policy evaluation that treats these parameters as known rather than uncertain produces overconfidence bias in its program effect estimates. Future work should apply this uncertainty propagation approach to the polygraph and iBorderCtrl parameters; Chat Control serves as the demonstration case because the net harm result is robust even under point estimate assumptions that are intentionally extremely generous to program proponents, making the propagation exercise conservative rather than determinative.

The structural equations in §4 were implemented as a running simulation (available at <https://github.com/verawilde/equilibrium-screening/tree/main/security>); §5.4 reports the four-pathway numeric results and propagates parameter uncertainty to the aggregate net effect Y^* by Monte Carlo ($P(\text{net harm}) = 100\%$ across 10,000 draws).

5.3 Four-pathway analysis

Applying the four-pathway framework to Chat Control 2.0, harm dominates along every causal pathway.

Classification: 99.4% of flags are false positives

Strategy: The net deterrence effect is likely near zero or negative for child safety outcomes regardless of deterrence magnitude. Dedicated offenders, who cause disproportionate harm, already use evasion tutorials on specialized Dark Net forums and will go dark.

Among casual offenders, deterrability is heterogeneous: most of those unaware of illegality cannot be deterred. Their detection is a classification event, not a strategy event. Some vaguely-aware casual offenders may be deterrable through public awareness campaigns. However, harm-weighted disaggregation shows that, even under generous assumptions about this deterrable fraction, the dedicated-offender evasion term dominates the aggregate strategy effect. We did not find a parameter combination under which the strategy pathway produces a net positive contribution to Y^* for Chat Control.

Sensitivity analysis over the deterrable fraction of casual offenders ($p_aware_casual \in [0, 1]$) shows that the harm-weighted strategy pathway effect is negative across the full parameter range (see Figure 3, §3.6), with dedicated-offender evasion exceeding maximum casual-offender deterrence by a factor of 46.6.

Information: No interrogation component; no bogus pipeline mechanism; no elicitation $\rightarrow \approx 0$

Resource reallocation: ~1.5B false positives overwhelm ~thousands of investigators $\rightarrow$ Catastrophic

5.4 Four-pathway simulation: numeric results

The structural equations in §4 are implemented as a Python simulation (available at <https://github.com/verawilde/equilibrium-screening/tree/main/security>). Key results are summarized here.

Literature sources for all parameters are collected in Appendix A.

In plain terms: across every parameter combination tested, Chat Control produces net harm; the police polygraph is unsigned — the simulation shows a modest net benefit that hinges entirely on a fragile information-pathway assumption, while the field data point the other way; and iBorderCtrl backfires under rarity even on its developers' own contested best-case accuracy, consistent with its abandonment.

Chat Control ($\pi = 1/1,000$, favorable assumptions): $Y^* = -5.06 \rightarrow$ net harm across all base rate values tested (1/100 to 1/10,000). Resource load $2.2\times$ investigative capacity. Under 10,000 Monte Carlo draws varying sensitivity, specificity, evasion rate, and base rate simultaneously: **P(net harm) = 100%**.

Police Polygraph ($\pi = 1/10$): $Y^* = +0.12 \rightarrow$ modest net benefit under baseline parameters, driven primarily by the information pathway (bogus pipeline, $d = .41$) offsetting negative classification and modest strategy effects.

This +0.12 is robust to base-rate uncertainty — a sensitivity analysis over π from 1/5 to 1/1,000 yields net benefit throughout, since strategy and information contributions are independent of π in this model — but it is conditional on the simulation's benign-information assumption that the bogus pipeline only screens problematic applicants out. It is the fragile result in the analysis: if the laboratory bogus pipeline effect ($d = .41$; Roese & Jamieson 1993) does not transport to operational

screening, runs in reverse via conscientiousness self-selection (§6.2, H3), or is outweighed by non-random false negatives selecting problematic recruits in (§6.2, H5), the sign flips. Simulation and empirical estimate (§6.1) disagree precisely here (see §6.4). Future work should propagate this transportability uncertainty formally (Gelman & Carpenter 2020).

To assess whether the net harm finding depends on the analyst's normative weighting of pathway contributions, we conducted a sensitivity analysis over theta values, varying θ_C (benefit per true positive: 0.5, 1.0, 2.0), θ_{FP} (harm per false positive: -0.1, -0.5, -1.0), and θ_R (resource overload penalty: -0.5, -2.0, -3.0) across all 27 combinations. θ_S and θ_I were not varied, as Chat Control's near-zero strategy effect and zero information yield mean these weights cannot affect the sign of Y^* .

Result: Y^* is negative in all 27 combinations, ranging from -7.36 to -1.66. The variation is driven almost entirely by θ_{FP} and θ_R ; θ_C is nearly inert, because under rarity true positives are too few to move the result. Even under the most favorable normative assumptions — minimal weight on false positive harm and resource costs — the program produces net harm. The net harm finding appears to be determined along this pathway by the program's structural properties.

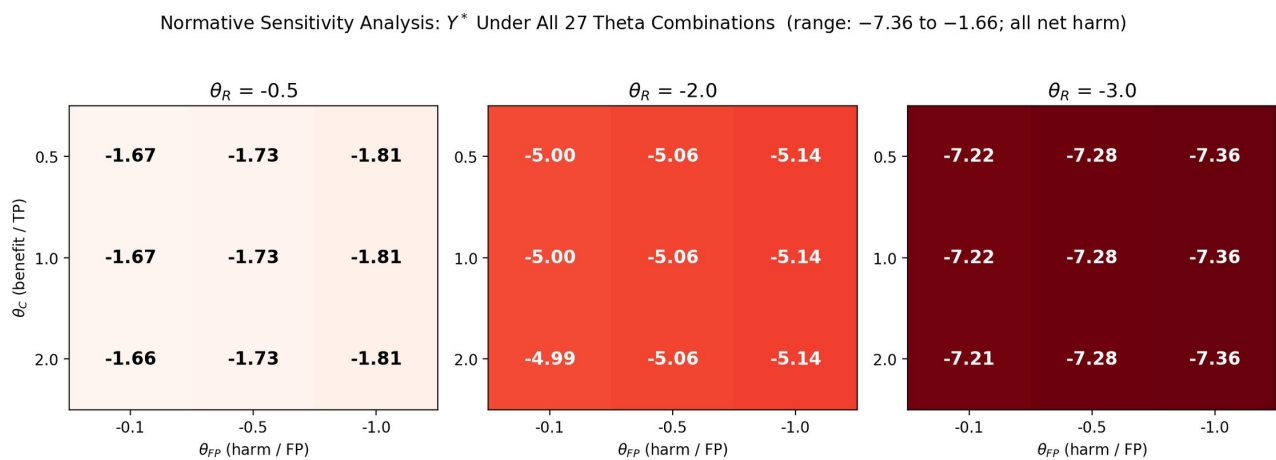


Figure 4. *Normative Sensitivity Analysis: Y^* Under All 27 Theta Combinations.* Each cell shows Y^* under a specific combination of θ_C (benefit per true positive: 0.5, 1.0, 2.0), θ_{FP} (harm per false

positive: -0.1 , -0.5 , -1.0), and θ_R (resource overload penalty: -0.5 , -2.0 , -3.0). θ_S and θ_I not varied (Chat Control strategy ≈ 0 , information = 0). (range: -7.36 to -1.66). Net harm appears structurally determined, robust to different normative weightings.

The resource-pathway result rests on capacity analysis detailed in the Supporting Information:

Europol IOCTA (2023) estimates approximately 3,000 law enforcement officers across EU member states engaged in CSAM-related work, and even assuming only 30 seconds of triage per flag, 1.598 billion false positive flags per 10 billion messages represent roughly 13.3 million person-hours — approximately 2.2 times total available capacity. The formal statement (Proposition 1 and Corollary 1), the institutional architecture, and the full calculations appear in Supporting Information S1. The 3,000-figure spans both layers. The specialist layer is approximately one order of magnitude smaller: scaling ZAC NRW's 13 prosecutors to the EU population suggests roughly 325 specialist prosecutors EU-wide, though such scaling should be treated as a placeholder given institutional heterogeneity across member states. Disaggregating the two layers shows the robustness argument is conservative, not generous. Even if triage is fully outsourced to non-specialist labor — scaling the first-pass layer indefinitely — the specialist investigation layer remains a binding constraint. At a 1% escalation rate from triage to specialist review (already very low), Chat Control generates ~ 16 million specialist-level cases per year against a capacity of 325,000: a $49\times$ overload. At 5%, the overload is $246\times$.

Coping with the workload requires clearing two sequential bottlenecks that cannot be resolved by hiring more triage workers. First, national police forensic investigators must conduct full forensic analysis of every case forwarded from the EU Centre; there is no lightweight substitute for this step under current law (Hartmann, expert interview, March 6, 2026). This presents the earlier and more acute constraint. Second, even if national police forensic capacity were somehow expanded, specialist prosecutors face a structurally different supply constraint: the pipeline of qualified CSAM

prosecutors is determined by legal training markets and professional factors, not funding availability. Both bottlenecks must be cleared simultaneously for Chat Control's output to be processed. Neither can be.

5.5 Chat Control equilibrium prediction

The four-pathway analysis converges on a single prediction: Chat Control 2.0 would net harm child safety.

Each pathway contributes to this conclusion independently. Along the classification pathway, applying Bayes' Rule produces estimated hypothetical outcomes in which 99.4% of flags are innocent even under assumptions intentionally favorable to proponents (see §5.1–5.2). Along the strategy pathway, the program fails to deter the offenders who cause disproportionate harm: dedicated criminals already use evasion tools and will migrate to darker infrastructure, while the subset of casual offenders who are genuinely unaware of illegality cannot be deterred by a program they do not know exists or applies to them. Even under maximally generous assumptions about casual-offender deterrability, dedicated-offender evasion dominates the aggregate strategy effect by a factor of 46.6 (§3.6, Figure 3). The information pathway contributes nothing: automated scanning generates no human interaction, no interrogation, no bogus pipeline mechanism. There is no elicitation. And along the resource reallocation pathway, the false positive flood overwhelms investigative capacity at every layer of the proposed architecture. This diverts finite resources away from the targeted investigations of known networks that currently constitute the most effective child protection work.

Unlike police polygraph programs, where strategy and information pathway contributions may compensate for classification pathway failure, Chat Control has no apparent logically compensating

possible mechanisms. The strategic and informational effects that could theoretically offset classification backfire are architecturally absent. All four pathways point in the same direction.

Prediction: Net harm to child safety.

6. APPLICATION 2: POLICE POLYGRAPH PROGRAMS

The empirical question is whether polygraph adoption at the department level predicts downstream workforce behavior. Agency data sought through earlier Freedom of Information Act requests for related research were withheld; this analysis therefore uses publicly available LEMAS data (Wilde 2016).

This section operationalizes that as citizen complaints of excessive use of force, a proxy for the kind of officer misconduct polygraph screening intends to select against.

6.1 LEMAS difference-in-differences findings

From U.S. nationally representative Law Enforcement Management and Administrative Statistics (LEMAS) data, 2003-2007, using difference-in-differences with negative binomial regression:

Sustained complaints (use of force):

- IRR: 3.0
- 95% CI .76-11.8
- $p = 0.12$

Total complaints (use of force):

- IRR: 2.4

- 95% CI 1.17-4.97
- $p = 0.02$

The analysis uses LEMAS 2003 (ICPSR 4411) and 2007 (ICPSR 31161), the only adjacent waves containing both screening tool usage and use-of-force complaint disposition counts. LEMAS 2016 (ICPSR 37323) dropped complaint count variables entirely. LEMAS 2020 (ICPSR 38651) restored them, but the 13-year gap between 2007 and 2020 spans major structural changes in American policing — post-Ferguson reforms, body camera adoption, consent decrees — making the parallel trends assumption indefensible. The hypothesized workforce selection mechanism would also be diluted by roughly two full cycles of officer turnover; we would generally expect a causal effect of an intervention to attenuate substantially over this amount of time.

Sustained citizen complaints of excessive use of force are rare events: most departments report none in any given survey wave (64% in 2003, 83% in 2007; see Figure 5), making the outcome a zero-inflated count with substantial overdispersion (Pearson $\chi^2/\text{df} = 113$). Negative binomial regression is appropriate for this count structure; results are reported as incidence rate ratios (IRR). The treatment group consists of departments that adopted polygraph screening between survey waves ($N \approx 33$ with complete covariate data); the control group consists of departments that never used polygraph screening in either wave. Covariates were log sworn officers and log population served (log-transformed to account for right skew in department size), collective bargaining authorization, internal affairs unit, and Census region.

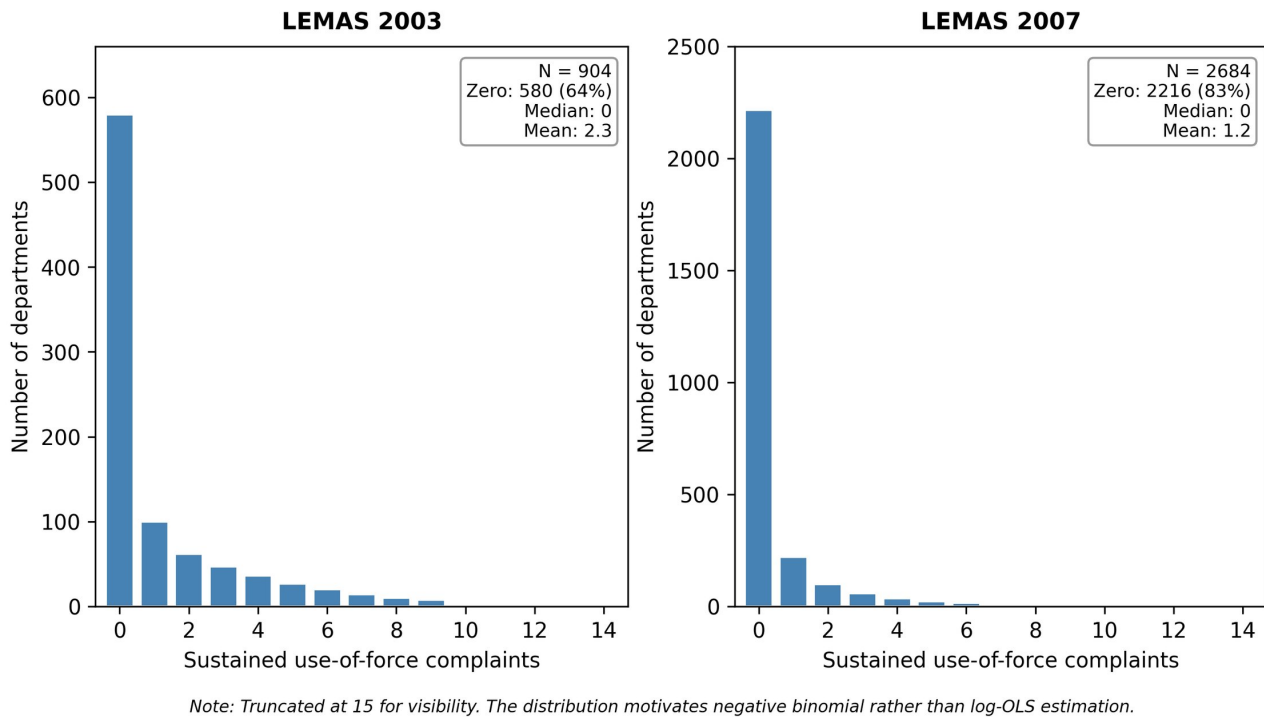


Figure 5. *Distribution of Sustained Use-of-Force Complaints per Department.* The severe zero-inflation, with 64% of departments reporting zero sustained complaints in 2003 and 83% in 2007, motivates negative binomial rather than log-OLS estimation. Truncated at 15 for visibility.

The 95% CI (0.76-11.8) is consistent with as much as a 24% decrease (IRR = 0.76) or an up to 1,080% increase (IRR = 11.8) in sustained complaints, reflecting uncertainty about both direction and magnitude. With a treatment group of $N \approx 33$, the estimate is too unstable to support conclusions about direction. This is consistent with a possible selection-for-non-confessors mechanism operating in reverse: polygraph screening may select officers who are less forthcoming under interrogation, and if sustained complaints require admissions or strong corroboration, effects in either direction are plausible depending on how complaint investigation processes interact with officer characteristics.

Total complaints show IRR = 2.4 (95% CI 1.17–4.97, $p = 0.02$), consistent with as little as a 17% increase (IRR = 1.17) or as much as a 397% increase (IRR = 4.97) in total complaints. Unlike the

sustained complaints compatibility interval, which includes both directions, this interval falls entirely above the null: the data are overall consistent with an increase in total complaints — more complaints, not fewer — though the magnitude is highly uncertain. Point estimates for sustained and total complaints trend in the same direction; but compatibility intervals are wide, N is small, and divergent effects remain compatible with the data. The sustained/total pattern that would most clearly support a non-classification pathway interpretation — fewer sustained complaints alongside unchanged or higher total complaints — can neither be proven nor disproven on the basis of this evidence.

6.2 Interpretation through four-pathway lens

The LEMAS data cannot isolate any single causal pathway, and so cannot identify which mechanism dominates police polygraph program net effects; fuller assessment would require mechanism-level implementation data (e.g., independent coders assessing polygraph confessions). What the four-pathway framework can do is structure the space of competing hypotheses and make explicit what each predicts. The empirical anchor for that exercise is the total use-of-force complaint result: $IRR = 2.4$, 95% compatibility interval 1.17–4.97 — an interval lying entirely above the null, consistent with an increase in complaints in adopting departments. Each hypothesis below is stated together with the direction it predicts for that outcome.

Hypothesis 1 (Classification/detection — benign; a benefit reading → predicts a decrease).

Polygraph testing detects problematic applicants via physiological measurement and screens them out, improving workforce quality. This predicts fewer complaints in adopting departments — a decrease. The observed increase does not support it. The compatibility interval is wide, however, and a true classification benefit cannot be ruled out: if troubled departments selectively adopt polygraph screening and remain troubled for reasons unrelated to the program, common-cause confounding could mask a real benefit behind a spurious positive association.

Hypothesis 2 (Strategy/deterrence — a benefit reading → predicts a decrease). Polygraph

screening deters problematic applicants from applying, or prompts them to self-select out during the process. This too predicts fewer complaints — a decrease. The observed increase does not support it, subject to the same wide-interval caveat.

Hypothesis 3 (Information/bogus pipeline — a benefit reading → predicts a decrease, but may

run in reverse → an increase). The interview/interrogation, rather than the physiological

measurement, does the work: the questioning process elicits admissions that screen out problematic applicants regardless of chart accuracy. This is the mechanism the §5.4 simulation leans on, and the

strongest theoretical basis for a program benefit (see §6.4). It also predicts fewer complaints — a

decrease. The observed increase does not support it, however. In fact, the observed increase

suggests the pathway could run the other way: a credible exam may elicit trivial or irrelevant

admissions disproportionately from conscientious applicants, who then screen out — draining the

workforce of its most conscientious recruits and raising misconduct. On this reading the information

pathway backfires, turning a presumed benefit into a harm.

Hypothesis 4 (Documentation artifact — an identification confound, not a causal pathway;

predicts an increase, program inert). The program does nothing, and the rise is an artifact of contemporaneous changes in recording.

Hypothesis 5 (Adversarial selection / non-random false negatives — a harm reading →

predicts an increase; program causally harmful). The polygraph's false negatives may not be

random noise. Deceptive subjects with lower socialization scores are more likely to pass the

examination (Waid, Orne & Wilson 1979), because low affective reactivity dampens the

physiological signal the test depends on. If — as this reading conjectures — that same low-

socialization, low-reactivity profile also predicts a propensity for abusive policing, then the test does

not merely fail to screen problematic recruits out; it selects them in, admitting the applicants most

prone to misconduct. This predicts more actual misconduct, and therefore an increase in complaints, with the program causally harmful. The conjectured socialization–misconduct link is itself an empirical claim requiring mechanism-level data to test.

These five hypotheses do not point the same way. Three of them — detection, deterrence, and bogus pipeline (H1–H3) — are the benefit readings, and all three predict a decrease in complaints. Only the two non-benefit readings predict an increase: the artifact story (H4), under which the program is inert, and the adversarial-selection story (H5), under which it is harmful. The empirical finding is an increase. The directional evidence is thus inconsistent with all three benefit hypotheses and consistent only with artifact or harm; on its face, the data lean against benefit.

Two cautions forbid converting that lean into a signed harm conclusion. First, complaints are a contaminated proxy: they conflate actual misconduct with the propensity to document it. That contamination is precisely the H4/H5 ambiguity — the same data are consistent with an inert program (more recording) and with a harmful one (more misconduct), and cannot separate them. Second, the treatment group is small ($N \approx 33$), the compatibility interval wide, and adoption non-random; as under H1's caveat, troubled departments may select into screening and remain troubled, so some or all of the observed rise may reflect pre-existing common causes rather than any program effect. The honest reading is that the directional evidence leans away from benefit and toward harm-or-artifact, while the net effect itself remains unsigned.

There are in fact two distinct artifact stories, both leaving the program causally inert: contemporaneous documentation tightening (H4), and non-random adoption — higher-scrutiny or more troubled departments selecting into screening (a between-department confound). Neither is a causal pathway; both can manufacture the observed increase.

Methods note: Log-transformed OLS is common in the policing literature for complaint outcomes (e.g., Rydberg and Terrill 2010; Terrill and Ingram 2016). But it is unreliable for zero-inflated

counts such as these (64–83% zeros; see Weisburd et al. 2022). Under these conditions, it can produce large coefficients that reflect distributional artifacts rather than treatment effects (Cameron and Trivedi 2013). Negative binomial regression, which models the count-generating process directly, is more appropriate for overdispersed count data (Cameron and Trivedi 2013; Houglund and Wolf 2016; Shjarback et al. 2021). It models the count-generating process directly and is used here for this reason.

6.3 Four-pathway analysis

Classification: Test validity is contested, and "inconclusives" are minimized in practice. The pathway is two-signed. Random classification error leaves its contribution weak and uncertain. But non-random false negatives — deceptive low-socialization recruits passing because low reactivity suppresses the physiological signal (Waid, Orne & Wilson 1979), with that same profile conjectured to predict abusive policing — would make the classification pathway actively harmful, selecting for problematic recruits rather than against them. LEMAS data cannot isolate which obtains.

Strategy: Deterrence of problematic applicants and self-selection out of the process are plausible benign mechanisms, but the LEMAS point estimate (an increase in total complaints) does not support a net deterrence benefit. The wide compatibility interval and absence of mechanism-level data leave the contribution uncertain.

Information: The bogus pipeline effect ($d = .41$; Roese & Jamieson 1993) is the strongest theoretical basis for a program benefit — confession elicitation screening out problematic applicants regardless of chart accuracy. But theory and data diverge here: confession-based screening-out predicts fewer complaints, whereas the difference-in-differences finds more. Either the laboratory bogus pipeline effect does not transport to operational applicant screening, or it is present but outweighed by other pathways.

Resource reallocation: Primary screening cost is approximately \$200–500/applicant and does not crowd out patrol resources. Secondary costs of false positives — additional testing, administrative appeals, legal challenges, employment costs for wrongly flagged applicants — are unquantified and could be substantial given the wide specificity range (.50–.80; NAS 2003). The pathway is unlikely to dominate the outcome but cannot be dismissed without department-level false positive estimates.

Across the four pathways, the theoretical case for benefit rests almost entirely on the information pathway, and the directional empirical evidence runs against it: total complaints rose, the opposite of what every benefit mechanism predicts. The pathways do not converge. The data lean toward harm or artifact, the simulation (§6.4) toward modest benefit, and the contradiction is left standing rather than resolved — the net effect cannot be signed without mechanism-level data.

6.4 Polygraph equilibrium prediction

Two lines of evidence point in opposite directions, and the honest result is their disagreement rather than either one alone.

The simulation (§5.4) predicts $Y^* = +0.12$ under baseline parameters — a modest net benefit, driven almost entirely by the information pathway (the bogus pipeline effect, $d = .41$). This dependence can be made precise. Re-running the simulation with a transport coefficient $\lambda \in [-1, 1]$ on the information pathway — $\lambda = 1$ if the laboratory bogus-pipeline effect transports fully to field screening, $\lambda = 0$ if it does not transport, $\lambda < 0$ if it inverts via adverse selection — shows that Y^* crosses zero at $\lambda^* \approx 0.30$. The modest net benefit holds only for $\lambda > \lambda^*$; at $\lambda = 0$ the program already nets harm ($Y^* = -0.05$). The LEMAS difference-in-differences, in which total complaints rise after adoption, is the empirical signature of λ below λ^* . The simulation therefore does not sign the net effect; it localizes the simulation–data disagreement to a single identifiable quantity — the field

transport of the bogus-pipeline effect — and the empirical direction points to the harm-or-inert branch.

The empirical difference-in-differences points the other way: total use-of-force complaints rose in adopting departments (IRR = 2.4, 95% compatibility interval 1.17–4.97, entirely above the null) — the opposite of what every benefit mechanism, including the bogus pipeline the simulation relies on, predicts.

This disagreement is reported, not reconciled. It is also informative, because it localizes the fragile assumption. The simulation's benefit depends on the bogus pipeline effect transporting from the laboratory settings of Roese & Jamieson (1993) to operational police applicant screening; if it does not transport, transports in the inverse along the information pathway, or is outweighed by adversarial selection through non-random false negatives along the classification pathway (§6.2, H5) — then the simulation's sign flips. The empirical estimate cannot adjudicate this: a small treatment group ($N \approx 33$), a wide compatibility interval, non-random adoption, and an outcome proxy that conflates misconduct with documentation all leave the net effect unsigned. The directional evidence leans against benefit and suggests possible negligible to modest harm; it does not establish practically important harm.

The polygraph case thus resolves neither way. It illustrates the limit of the framework as an empirical tool in the security domain, where ground truth is unobservable — and, equally, the value of surfacing the contradiction: it identifies exactly what future mechanism-level data must settle, namely whether the bogus pipeline operates pro or con in operational screening and whether false negatives are non-random or systematic in potentially deleterious ways (e.g., selection against conscientious recruits). This contrasts with Chat Control, which converges on harm across all four pathways regardless of such decomposition; polygraph program efficacy cannot be evaluated without it. The framework's contribution in this case is therefore not a verdict but legibility: it

makes the disagreement between simulation and data explicit and locates which mechanism is load-bearing, rather than letting a single mis-signed estimate stand in for an answer.

7. APPLICATION 3: iBORDERCTRL (EU BORDER SECURITY)

iBorderCtrl was an EU Horizon 2020-funded pilot (2016-2019) piloting AI “lie detection” at non-Schengen border crossings in Hungary, Latvia, and Greece (Roussos et al., 2019). The system used facial micro-expression analysis based on Ekman and Friesen’s (1978) action coding system to flag travelers for secondary screening, despite ample evidence that inferring emotional states or deceptive intent from facial movements lacks adequate cross-cultural validation (e.g., Barrett et al., 2019) and is otherwise problematic in signal detection terms (NAS, 2003). iBorderCtrl offered, in effect, an interrogation with no interrogator: it contained the form of the information pathway — a kiosk ‘interview’ — without the adaptive human elicitation that gives the bogus pipeline its potential efficacy.

Four-pathway analysis:

Classification: No validated accuracy data; AI emotion detection lacks adequate scientific evidentiary basis → Unknown; illustratively, backfires under rarity according to developers’ own figures

Strategy: Sophisticated bad actors (smugglers, traffickers) likely evade; naive travelers flagged → Negative

Information: Some bogus pipeline potential in interviews, but automated kiosk format limits elicitation → Weak

Resource reallocation: Flags require human follow-up; unknown false positive burden →
Uncertain

Illustrative simulation. No validated accuracy data exist for iBorderCtrl due to non-transparency. Indeed, iBorderCtrl is itself a case study in the transparency problem this framework identifies. When MEP Patrick Breyer sought access to the project's documentation under EU freedom of information rules, the Research Executive Agency withheld material bearing on the system's reliability, false positive performance, ethics, and legality on commercial interest grounds — a refusal the EU General Court (Case T-158/19, 15 December 2021) substantially sustained and the Court of Justice upheld on appeal (Case C-135/22 P, 7 September 2023). The pilot study's own performance evidence therefore remains sealed; the figures used below come from academic publications on the system on which iBorderCtrl's "AI lie detector" was based (O'Shea et al. 2018), not from the withheld pilot results. In addition, the developers' headline 76% figure does not represent a single-construct accuracy rate: it amalgamates hit rates across distinct emotion categories and across error types, so it cannot be read as a validated deception detection accuracy figure (O'Shea et al. 2018).

Following the generous assumptions logic applied to Chat Control in §5.1, we can nonetheless grant the developers their own contested lab figures — roughly 73.7% sensitivity and 75.6% specificity from a 30-participant validation (O'Shea et al. 2018) — and ask what follows.

Under any plausible border base rate the classification pathway backfires: PPV falls to ~3% at $\pi = 1/100$ and ~0.3% at $\pi = 1/1,000$ (97–99.7% of flags innocent), a worse false positive burden than Chat Control because the claimed specificity (0.76) is below Chat Control's assumed 0.84. Running the full four-pathway simulation on these figures, and generously granting that the kiosk's automated elicitation functions as a benefit ($\lambda = 1$), still yields net harm ($Y^* \approx -0.35$), because

sophisticated-actor evasion and classification backfire dominate the modest information benefit; with no information benefit ($\lambda = 0$), $Y^* \approx -0.52$.

Because resource load is negligible at pilot scale, this result is insensitive to the assumed screening volume; the harm is classification- and strategy-driven. These figures are illustrative only; they rest on the developers' own unvalidated numbers and an assumed base rate. They are offered to show the program fails even on its proponents' best case, not to estimate its true effect.

Outcome: The research project faced substantial scientific and civil liberties criticism, and was not adopted for operational use. It is a qualified success case for critique — but the qualification matters. What foreclosed deployment was not public evidence that the system failed; the developers' own reliability data were never disclosed (see below). Rather, it was the independently established implausibility of the underlying science (Barrett et al. 2019), civil society and scholarly mobilization, and the transparency litigation itself. iBorderCtrl is thus both a success case for critique and an illustration of this paper's broader point that non-transparency often blocks scientific evidentiary evaluation of programs of this structure. iBorderCtrl was stopped despite, not because of, access to the data needed to evaluate its evidentiary basis.

8. THE BROADER CLASS: PROGRAMS SHARING THIS STRUCTURE

As discussed in the introduction, mass screenings for low-prevalence problems recur across otherwise unrelated domains:

Security: Population polygraph screening, Chat Control 2.0, iBorderCtrl, terrorist watchlists, airport security

Health: Asymptomatic cancer screening (mammography, PSA, colonoscopy), prenatal testing, newborn screening

Information integrity: Plagiarism detection, AI content detection, misinformation flagging, disinformation detection

Education: Academic integrity monitoring, proctoring AI

Finance: Fraud detection, AML screening

All share an identical mathematical structure as signal detection problems, and face the same accuracy-error tradeoff which leads them to risk incurring invisible net societal damages under conditions of rarity, persistent uncertainty, and secondary screening harms. Yet, they are rarely recognized as a class requiring common analytical treatment.

A further irony: law enforcement agencies themselves rely on encrypted communications for operational security. A Chat Control architecture scanning all digital communications would sweep up their own, a fact noted by practitioners as an important operational concern in addition to the broader liberal democratic one (Hartmann, expert interview, May 21, 2025).

Policy implication: Rather than fighting these programs case-by-case (Whack-A-Mole), society needs to legislate the structure — requiring evidence-based evaluation of programs sharing this dangerous mathematical form before implementation.

Two cases illustrate that critique can succeed. Fienberg's Congressional testimony on the NAS report halted the Department of Energy's proposed mass polygraph screening of National Lab scientists: a success grounded in a public, authoritative evidentiary review. Similarly, iBorderCtrl was abandoned after sustained scientific and civil liberties criticism. In both cases, the data that might have permitted independent analyses of program efficacy and fairness were withheld despite concerted efforts to obtain them under Freedom of Information rules. Critics' success turned on the implausibility of the method and on transparency- and rights-based mobilization rather than

disclosed performance data. Programs of this structure should be built to be evaluable, with their evidence public, so critique need not proceed in the dark and massive net societal harm cannot invisibly accumulate.

9. COMPARATIVE ANALYSIS

Scale

- Chat Control: ~450M people, billions of messages
- Police Polygraph: ~thousands of applicants per year
- iBorderCtrl: ~thousands of travelers per day at pilot sites

Base rate

- Chat Control: Extremely low (~0.001%); unverified — industry-provided figures, independent estimates unavailable (see §5.2). Here as across cases, there has been debate about what exactly the problem of interest is (e.g., CSAM transmission versus grooming).
- Police Polygraph: Higher (~5–20%), highly uncertain; no validated prevalence estimate for ‘problematic applicants’ exists. Range is illustrative and the category again contested (e.g., violent criminals versus other categories of possible offenders).
- iBorderCtrl: Low, unknown — no published base rate estimate for smugglers/traffickers as fraction of border crossers. Again the category could be construed in quite different ways (e.g., people who might intend to cross the border and see if they could stay longer than planned versus terrorists).

Test validity

- Chat Control: AI accuracy inflated/unverifiable
- Police Polygraph: Contested but irrelevant if Information pathway dominates
- iBorderCtrl: No validated accuracy data; pilot reliability data withheld on commercial-interest grounds (Breyer v REA, 2021/2023); AI emotion detection lacks adequate scientific evidentiary basis

Selection effects

- Chat Control: Dedicated offenders migrate to encrypted channels
- Police Polygraph: Problematic applicants self-select out of the process
- iBorderCtrl: Sophisticated actors choose alternative crossings.

Information pathway

- Chat Control: Zero (automated, no human interaction)
- Police Polygraph: Strong (interrogation, bogus pipeline)
- iBorderCtrl: Weak — automated avatar had some adaptive questioning capacity, but the depth of possible automated adaptive elicitation is far shallower than that of human interview/interrogation, potentially limiting bogus pipeline potential

Strategy pathway

- Chat Control: Evasion dominates for sophisticated offenders
- Police Polygraph: Deterrence may dominate for applicant pool

- iBorderCtrl: Sophisticated actors evade; naive travelers flagged

Resource impact

- Chat Control: Catastrophic (billions of false positives)
- Police Polygraph: Manageable (\$200-500/applicant)
- iBorderCtrl: Uncertain (unknown false positive burden at scale)

Feedback dynamics

- Chat Control: Victim advocacy (Heat Initiative) sustains support despite poor evidence
- Police Polygraph: Institutional inertia; “lie detector” mystique in American context
- iBorderCtrl: EU funding cycles sustained pilot despite widespread scientific and social critique

Predicted net effect

- Chat Control: Harm
- Police Polygraph: Unsigned. Simulation predicts modest benefit via the information pathway; the LEMAS difference-in-differences points the other way (total complaints up). Disagreement unresolved; mechanism-level data needed.
- iBorderCtrl: Likely harm; abandoned after critique

10. SIMULATION STRATEGY

The four-pathway model is implemented as the running simulation reported in §5.4 (code and data at <https://github.com/verawilde/equilibrium-screening/tree/main/security>). Three extensions would strengthen it. First, recasting the cyclic structural model as a deep-equilibrium layer solved by Anderson acceleration (Besserve and Schölkopf 2022), with deflated Newton methods (Farrell et al. 2015) to surface multiple equilibria and the tipping points at which feedback loops flip sign. Second, full Bayesian propagation of parameter uncertainty across all pathways — including the transportability uncertainty flagged in §3.5, currently unquantified and likely larger than the within-study sampling error (Gelman and Carpenter 2020). Third, multi-level structural equations carrying the within-pathway heterogeneity the aggregation problem requires (§3.6).

The highest-value empirical extension is estimating the deterrable fraction of casual offenders ($p_{\text{aware_casual}}$), the parameter to which the strategy-pathway disaggregation is most sensitive in principle. Jurisdictions where public-awareness campaigns accompanied high-profile CSAM prosecutions without mandatory scanning could supply before/after variation identifying its upper bound independently of classification-pathway effects; platform-level data on report volumes around such campaigns would sharpen the estimate further. Such an estimate would replace the present qualitative foreclosure of the compensating-subpopulation argument (§3.6) with an empirically anchored one.

11. CONCLUSION

This paper makes four contributions. First, it extends the National Academy of Sciences' single-pathway analysis of mass screening programs (NAS 2003) to a four-pathway causal framework encompassing classification, strategy, information, and resource reallocation effects. The key theoretical insight is that programs sharing the mathematical structure of signal detection under rarity, persistent uncertainty, and secondary screening harm operate through multiple causal

channels simultaneously, and that the net effect cannot be determined from classification accuracy alone.

Second, it applies Besserve and Schölkopf's (2022) deep equilibrium framework to formalize the feedback loops through which screening programs reshape the populations they claim to characterize, an instance of Perdomo et al.'s (2020) performative prediction problem. The structural equations and simulation code (<https://github.com/verawilde/equilibrium-screening/tree/main/security>) provide a reusable toolkit for analyzing other programs in the broader class.

Third, the comparative analysis of Chat Control, police polygraph programs, and iBorderCtrl demonstrates the framework's diagnostic value. For Chat Control, all four pathways converge on net harm: classification backfires under Bayes' Rule, dedicated-offender evasion dominates the strategy pathway even under generous assumptions about casual-offender deterrence, and the information pathway is architecturally absent. iBorderCtrl occupies an intermediate position where weak coupling appears likely insufficient to net compensate for the Bayesian backfire and resource reallocation costs.

This convergence is robust to normative weighting of pathway contributions (§5.4) and to disaggregation of within-pathway heterogeneity (§3.6). The disaggregation decision rule was formulated because proponents could invoke a compensating subpopulation argument for Chat Control: the disaggregated simulation closes that argument empirically rather than by assumption.

For police polygraph programs, the pathways do not provably converge on a net effect of one sign or another. The simulation predicts modest net benefit via the information pathway, but the LEMAS difference-in-differences points the other way — total complaints rise in adopting departments — leaving simulation and data in unresolved disagreement; with a small treatment group and wide compatibility intervals, the net effect remains unsigned.

For iBorderCtrl, the absence of validated accuracy data — its performance evidence withheld on commercial-interest grounds and never disclosed (§7) — together with the absent information pathway, suggests net harm, consistent with the program’s abandonment after scientific and civil society critique.

Fourth, the paper identifies a structural regularity across apparently unrelated policy domains — security screening, medical screening, information integrity, education, and finance — and argues that the recurring policy failure of implementing programs that backfire under conditions of rarity, uncertainty, and secondary screening harm necessitates structural legislation rather than case-by-case advocacy.

The framework has important limitations (§3.5–3.6). It is better understood as a qualitative diagnostic tool than a quantitative predictive one, particularly in the security domain where ground truth is often persistently unobservable. The aggregation problem of collapsing within-pathway heterogeneity into single parameters is partially addressed by the suggested disaggregation decision rule and implemented simulations, but remains a persistent challenge. Future work should pursue cross-domain validation proceeding from medical applications, where ground truth is relatively potentially observable and many high-quality datasets accessible, toward security applications where they are not.

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APPENDIX A. PARAMETER ESTIMATES FROM LITERATURE

Bogus pipeline effect

- Estimate: $d = .41$
- 95% CI .25-.57
- Source: Roese & Jamieson (1993) meta-analysis

Deterrence (certainty — perceived likelihood of being caught)

- Estimate: $r \sim 0.15$
- Plausible range: [0.05, 0.25]
- Source: Pratt et al. (2006); Nagin (2013)

Deterrence (severity — perceived consequences if caught)

- Estimate: $r \sim 0$
- Plausible range: [-0.05, 0.05]
- Source: Doob & Webster (2003); Nagin (2013)

Polygraph sensitivity

- Estimate: 0.70-0.90
- Source: NAS (2003); varies substantially by method

Polygraph specificity

- Estimate: 0.50-0.80

- Source: NAS (2003)

CSAM AI accuracy

- Hash matching (known CSAM only): Industry claims ~99% accuracy, but only detects images already in database — cannot find new CSAM
- AI classifiers (new CSAM): Microsoft claims 88% for grooming detection; Thorn claims >99% for images — no independent verification exists
- Sources: European Commission (2023) citing Microsoft; Thorn marketing; Steinebach (2023)
- Note on targeted vs. mass screening accuracy: AI systems optimized for precision on seized targeted evidence — data already filtered by reasonable suspicion — show 5-7% error rates in that setting (Hartmann, expert interview, May 21, 2025). Mass screening on unfiltered general communications would be expected to perform substantially worse. The 90% true positive rate used in §5.1 is therefore generous even by the standards of optimized targeted tools.
- Note: Following Fienberg's generous-assumptions approach, our Bayesian analysis uses 90% accuracy. Even under these favorable conditions, the program backfires catastrophically due to base rate rarity.